

Galaxy for virologist training. Exercise 3: Illumina Assembly 101

Title	Galaxy
Training dataset:	PRJEB43037 - In August 2020, an outbreak of West Nile Virus affected 71 people with meningoencephalitis in Andalusia and 6 more people in Extremadura (south-west of Spain), causing a total of eight deaths. The virus belonged to the lineage 1 and was relatively similar to previous outbreaks occurred in the Mediterranean region. Here, we present a detailed analysis of the outbreak, including an extensive phylogenetic study. This is one of the outbreak samples.
Questions:	• What is an assembly? • How can I evaluate my assembly?
Objectives:	• Understand the assembly concept. • Learn how to interpret assembly quality control metrics.
Estimated time:	40 min

1. Description

Sometimes, we don't have a reference genome to map our reads against, or we want to reconstruct a genome without any possible bias caused by a reference. In such cases, we need to do a ***de novo assembly***. This type of analysis tries to reconstruct the original genome without any templates, using only the reads themselves. Some considerations to consider beforehand:

- When we assemble, the longer the reads are and the longer the size of the library fragments, the easier it is for the assembler. That's why PacBio or Nanopore are recommended for assembly. Think of it like a puzzle, the bigger the pieces, the easier it is to form the image.
- It's almost imposible to reconstruct the entire genome of a large-genome microorganism with only one sequencing, although it can be done for smaller ones, like viruses.
- Assembly is not recommended for amplicon based libraries due to the depth of coverage unevenness and the amplicons' intrinsic bias.

2. Upload data to Galaxy

Training dataset

- Experiment info: PRJEB43037, WGS, Illumina MiSeq, paired-end.
- Fastq R1: [ERR5310322_1](ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_1.fastq.gz) - URL :
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_1.fastq.gz

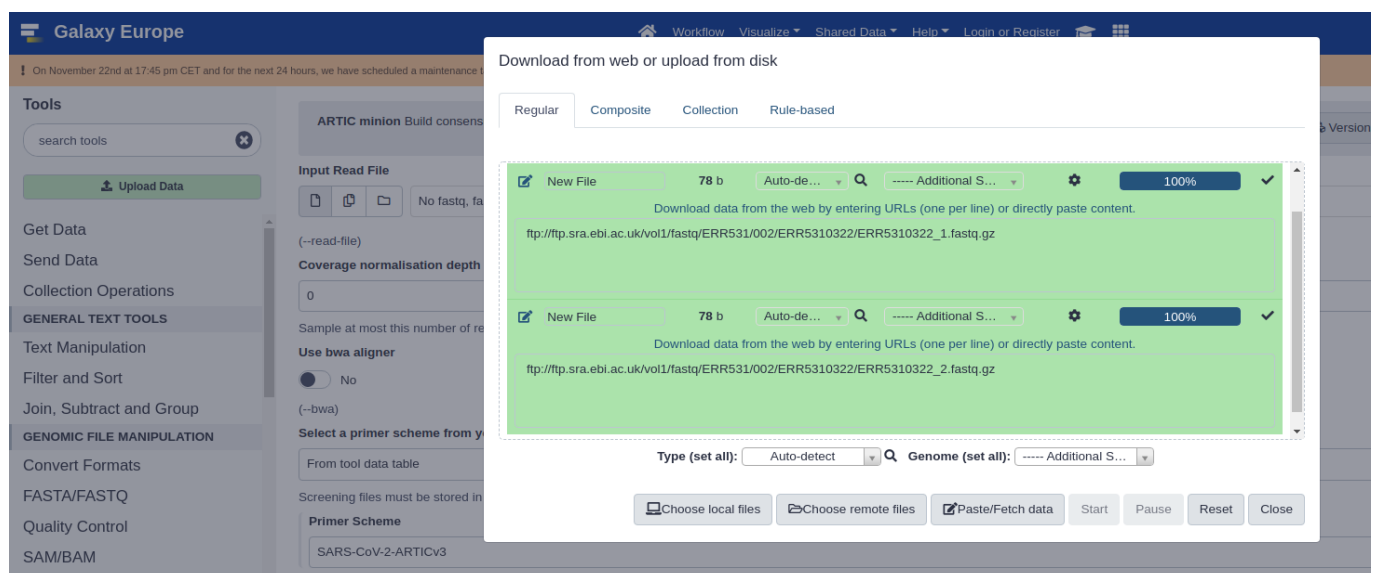
- Fastq R2: [ERR5310322_2](ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_2.fastq.gz) URL :
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_2.fastq.gz
- Reference genome **NC_009942.1**: **FASTA** -- **GFF**

Create new history

- Click the **+** icon at the top of the history panel and create a new history with the name **Illumina Assembly** as explained [here](#).

Upload data

- Import and rename the read files **ERR5310322_1** and **ERR5310322_2**
 - Click on **Upload**.
 - Click on **Paste/Fetch data**.
 - Copy URL for fastq R1 (select and Ctrl+C) and paste (Ctrl+V).
 - Click on **Start**.
 - Wait until the job finishes (green in history).
 - Do the same for fastq R2.



- Rename R1 and R2 files.**
 - Click on the in the history for **ERR5310322_1.fastq.gz**.
 - Change the name to **ERR5310322_1**.
 - Do the same for R2.

Name

Info

Annotation

Add an annotation or notes to a dataset; annotations are available when a history is viewed.

Database/Build

----- Additional Species Are Below -----

7: GCF_000875385.1_Viral
Proj30293_genomic.fna.gz

1 sequences
format: **fasta.gz**, database: ?

display with IGV local

```
>NC_009942.1 West Nile virus lineage 1, con
AGTAGTTCCGCTGTGTGAAGCTGACAACTTAGTAGTTTGTG
TAGCAGCAAGATCTCATGTTCTAAGAAACCAAGAGGCCCGC
CCGCGTGTGTCTCTGATTTGAGTGAAGAGGAGGCTATGTTGA
GCTCTCTTGGCGTCTTCAGGTTTCAGCAATTGCTCCGACCC
```

6: ERR5310322_2

5: ERR5310322_1

- Import the reference genome and GFF file.

https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/875/385/GCF_000875385.1_ViralProj30293/GCF_000875385.1_ViralProj30293_genomic.fna.gz
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/875/385/GCF_000875385.1_ViralProj30293/GCF_000875385.1_ViralProj30293_genomic.gff.gz

Descargar de la red o cargar desde disco

RegularCompositeCollectionRule-based

You added 1 file(s) to the queue. Add more files or click 'Start' to proceed.

Name	Size	Type	Genome	Settings	Status
New File	267 b	Auto-de...	unspecified (?)		0%

Download data from the web by entering URLs (one per line) or directly paste content.
pi.nlm.nih.gov/genomes/all/GCF/000/875/385/GCF_000875385.1_ViralProj30293/GCF_000875385.1_ViralProj30293_genomic.fna.gz
pi.nlm.nih.gov/genomes/all/GCF/000/875/385/GCF_000875385.1_ViralProj30293/GCF_000875385.1_ViralProj30293_genomic.gff.gz

Type (set all): Auto-detect Genome (set all): unspecified (?)

Elegir archivos locales

Choose remote files

Paste/Fetch data

Start

Pause

Reset

Close

- Rename the reference genome and gff file.
 1. Click on the for the reference file in the history.
 2. Change the name to **NC_009942.1**.

Name

GCF_000875385.1_ViralProj30293_genomic.fna.gz

Info

Annotation

Add an annotation or notes to a dataset; annotations are available when a history is viewed.

Database/Build

----- Additional Species Are Below -----

7: GCF_000875385.1_ViralProj30293_genomic.fna.gz

z

1 sequences

format: fasta.gz, database: ?

display with IGV local

>NC_009942.1 West Nile virus lineage 1, co
AGTAGTTCCGCTGTGTGAGCTGACAAAGCTTAAGTGT
TATGACAGAGATCTCGATGCTAAGAACCAAGAGGCCG
CCCCGCTGTTGCTCTTGAATGAGTGAAGAGGCTATGTTGA
GCTCTCTTGGCCTCTCTCAGGTTACAGCAATTGCTCCGACC

6: ERR5310322_2

5: ERR5310322_1

- Finally, add some useful tags if you like.

Assemble reads with SPAdes

1. Search **Spades** in the search tool box and select *rnaviralSPAdes de novo assembler for transcriptomes, metatranscriptomes and metaviromes*.

2. Single-end or paired-end short-reads > **Paired-end: list of dataset pairs.**

[!WARNING] To be able to continue, it is necessary to first create a **list of paired datasets**, given our .fastq files, which will then be used for the rest of the steps of this workshop. In order to do so, we must:

1. Select our fastqsanger.gz files that we uploaded in the beginning, and then click on *2 of X selected*.
2. Select *Advanced Build List*.
3. Select the *List of Paired Datasets* box and click on *Next*.
4. Check the auto-pairing configuration and click again on *Next*.
5. Give a name to your list and finally click on *Build*.

Once this is done, you'll be able to select your list as input of SPAdes.

3. FASTQ RNA-seq file(s): collection: **ERR5310322**
4. Select optional output file(s) > **Scaffolds stats** and **Contigs stats**, along with the options selected by default.
5. Click on **Run Tool** and wait.

Herramientas

spades 1 x

Cargar Datos

Show Sections

SPAdes genome assembler for genomes of regular and single-cell projects

metaviralSPAdes extract and assembly viral genomes from metagenomic data

metaplasmidSPAdes extract and assembly plasmids from metagenomic data

plasmidSPAdes extract and assembly plasmids from WGS data

metaSPAdes metagenome assembler

rnaviralSPAdes de novo assembler for transcriptomes, metatranscriptomes and metaviromes 2

biosyntheticSPAdes biosynthetic gene cluster assembly

coronaSPAdes SARS-CoV-2 de novo genome assembler

rnaSPAdes de novo transcriptome assembler

Shovill Faster SPAdes assembly of Illumina reads

rnaviralSPAdes de novo assembler for transcriptomes, metatranscriptomes and metaviromes (Galaxy Version 3.15.4+galaxy2)

Operation mode

Assembly and error correction

To run read error correction, reads should be in FASTQ format.

Single-end or paired-end short-reads

Paired-end: individual datasets

It assumes that all samples belong to the same library. If you want to use samples from two different libraries, include the second library as additional set of short-reads.

FASTQ RNA-seq file(s): forward reads

3: NC_009942.1
2: ERR5310322_2
1: ERR5310322_1

FASTQ RNA-seq file(s): reverse reads

3: NC_009942.1
2: ERR5310322_2
1: ERR5310322_1

Type of paired-reads

Default (--pe)

In paired-end sequencing, the library preparation yields a set of fragments, and the machine sequences each fragment from

History

buscar conjuntos de datos

illumina assembly 101 tutorial

31.4 MB 4 4

4: NC_009942.1 GFF #gff

3: NC_009942.1 #reference

2: ERR5310322_2 #reverse

1: ERR5310322_1 #forward

[!WARNING] **Assembly takes time!** There is no such thing as Assembly in real time. It can take anywhere between 90 minutes and two hours.

Click on the eye icon in the history: **Spades Contigs stats.**

- ## Assembly quality control with QUAST

1. Search QUASt in the search tool box.
2. **Assembly mode? > Individual assembly (1 contig file per sample).**
3. Contigs/scaffolds file > **rnaviralSpades Scaffolds**
4. Use a reference genome? > **Yes.** Select the **NC_009942.1 FASTA file** previously loaded.
5. Genomic feature positions in the reference genome > **NC_009942. GFF file** previously loaded.

Galaxy Europe Flujo de Trabajo Visualizar Datos Compartidos Ayuda Usuario

Herramientas

quast 1

Cargar Datos

Show Sections

Quast Genome assembly Quality 2

rnaQUAST A Quality Assessment Tool for De Novo Transcriptome Assemblies

FLUJOS DE TRABAJO

Todos los flujos de trabajo

Quast Genome assembly Quality (Galaxy Version 5.2.0+galaxy0)

Use customized names for the input files?

No, use dataset names

They will be used in reports, plots and logs

Contigs/scaffolds file

12: rnaviralSPAdes on data 2 and data 1: Scaffolds
11: rnaviralSPAdes on data 2 and data 1: Contigs
3: NC_009942.1 (as fasta)

3

Reads options

Disabled

Currently, the supported read types are Illumina unpaired, paired-end and mate-pair reads, PacBio SMRT, and Oxford Nanopore long reads.

Type of assembly

Genome

Use a reference genome?

Yes

4

Many metrics can't be evaluated without a reference. If this is omitted, QUAST will only report the metrics that can be evaluated without a reference.

Reference genome

12: rnaviralSPAdes on data 2 and data 1: Scaffolds
11: rnaviralSPAdes on data 2 and data 1: Contigs
3: NC_009942.1 (as fasta)

5

Galaxy Europe Flujo de Trabajo Visualizar Datos Compartidos Ayuda Usuario

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quast

Cargar Datos

Show Sections

Quast Genome assembly Quality

rnaQUAST A Quality Assessment Tool for De Novo Transcriptome Assemblies

FLUJOS DE TRABAJO

Todos los flujos de trabajo

Genomic feature positions in the reference genome

4: NC_009942.1 GFF (as bed)

6

Gene coordinates for the reference genome (--features)

Operon positions in the reference genome

Nothing selected

Operon coordinates for the reference genome (--operons)

Compute k-mer-based quality metrics?

No

It is recommended for large genomes. This may significantly increase memory and time consumption on large genomes (--k-mer-stats)

Generate Circos plot

No

Plot Circos version of Icarus contig alignment viewer (--circos)

7 Execute

6. Run Tool.

7. Click on the eye icon of the **QUAST HTML report**.

- ▶ How much of or reference genome have we reconstructed?
- ▶ How many contigs do we have greater than 1000 pb?
- ▶ How long is the largest contig in the assembly?
- ▶ Which is the N50?

8. Open the **Icarus** viewer in the quast report.

QUAST

Quality Assessment Tool for Genome Assemblies by [CAB](#)

17 November 2021, Wednesday, 18:07:45

[View in Icarus contig browser](#)

All statistics are based on contigs of size ≥ 500 bp, unless otherwise noted (e.g., "# cor bp)" and "Total length (≥ 0 bp)" include all contigs).

Aligned to "dataset_76f460df_9dce_4919_9108_be70c4d29af9" | 11 029 bp | 1 fragment
G+C

Genome statistics

SPAdes_on_data_3_and_data_2...

Genome fraction (%)

83.070

- How did the contig align against our reference genome?

This training history is available at: <https://usegalaxy.eu/u/s.varona/h/illumina-assembly-101-tutorial>